A Stylometric Application of Large Language Models

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Source: arXiv:2510.21958 • Code: https://github.com/ContextLab/llm-stylometry

We show that large language models (LLMs) can be used to distinguish the writings of different authors. Specifically, an individual GPT-2 model, trained from scratch on the works of one author, will predict held-out text from that author more accurately than held-out text from other authors. We suggest that, in this way, a model trained on one author's works embodies the unique writing style of that author. We first demonstrate our approach on books written by eight different (known) authors. We also use this approach to confirm R. P. Thompson's authorship of the well-studied 15th book of the Oz series, originally attributed to F. L. Baum.

1 Introduction

Herein we introduce *predictive comparison*, a new LLM-based relative stylometric measure. It derives from a simple idea, that if an LLM can be trained to write like—i.e., in the style of—a given author by training on their work [e.g., ?], then the degree to which such a model can be used to predict held-out text from any *candidate* author should reflect the relative similarity of the styles of the two authors.

For ease of exposition, throughout this paper we focus on prose written by known novelists. To this end, we adapted and trained a series of small GPT-2 models [?] to write in the style of any of $N = 8$ different authors. Specifically, for each author, we trained one model on a training partition of that author's complete works, and evaluated each model's predictive accuracy on held-out passages from each of the eight authors.

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Code availability

Code: `https://github.com/ContextLab/llm-stylometry`**. See the project's README for setup instructions.**

Note

This is an abbreviated re-styled excerpt of the full paper, produced as a demonstration of the llmXive house style. The full content (Methods, Results, Discussion, all figures, and the complete bibliography) is available in the original on arXiv at arXiv:2510.21958.